

ALGEBRAIC IDENTITIES

Expand & Factorise · Grade 6

What Are Algebraic Identities?

Algebraic identities are special equations that are true for every value of the variables. They act like shortcuts — instead of multiplying everything out by hand, you can recognise a pattern and jump straight to the answer!

Example: $(a + b)^2$ is always equal to $a^2 + 2ab + b^2$, no matter what a and b are.

The Four Key Identities

Label	Formula	Name
Identity 1	$(a + b)^2 = a^2 + 2ab + b^2$	Square of a Sum
Identity 2	$(a - b)^2 = a^2 - 2ab + b^2$	Square of a Difference
Identity 3	$a^2 - b^2 = (a + b)(a - b)$	Difference of Two Squares
Identity 4	$(a + b)(a + c) = a^2 + (b+c)a + bc$	Product of Two Binomials

Identity 1 : $(a + b)^2 = a^2 + 2ab + b^2$

Use this when you see **a binomial squared with a plus sign**. Remember: Square first · Double product · Square last.

Worked Example — Expand $(x + 3)^2$

Step 1 : Identify $a = x$ and $b = 3$

Step 2 : Write the formula $(a + b)^2 = a^2 + 2ab + b^2$

Step 3 : Substitute $x^2 + 2 \times x \times 3 + 3^2$

Step 4 : Simplify $\rightarrow x^2 + 6x + 9$

Tip: Square the first term + twice the product of both terms + square the last term.

Identity 2 : $(a - b)^2 = a^2 - 2ab + b^2$

Same structure as Identity 1 but the **middle term becomes negative**. The last term (b^2) is always positive because a negative squared is positive.

Worked Example — Expand $(y - 4)^2$

Step 1 : Identify $a = y$ and $b = 4$

Step 2 : Write the formula $(a - b)^2 = a^2 - 2ab + b^2$

Step 3 : $a^2 = y^2$

Step 4 : $2ab = 2 \times y \times 4 = 8y \Rightarrow$ with minus sign: $-8y$

Step 5 : $b^2 = 4^2 = 16$

Step 6 : Final answer $\rightarrow y^2 - 8y + 16$

Identity 3 : $a^2 - b^2 = (a + b)(a - b)$

Use this to **factorise** an expression that is a difference of two perfect squares. Look for two terms separated by a minus sign where each is a perfect square.

Worked Example — Factorise $9x^2 - 16$

Step 1 : Recognise perfect squares $9x^2 = (3x)^2$ and $16 = 4^2$

Step 2 : Apply the identity $a^2 - b^2 = (a + b)(a - b)$

Step 3 : Substitute $a = 3x$ and $b = 4$

Step 4 : Answer $\rightarrow (3x + 4)(3x - 4)$

Tip: Always check — can you write each term as something²? If yes, use Identity 3!

Identity 4 : $(a + b)(a + c) = a^2 + (b+c)a + bc$

Use this when expanding two binomials that **share a common variable term**. An area model (grid) makes it easy to visualise every product.

Worked Example — Expand $(x + 2)(x + 5)$

Step 1 : Identify $a = x$, $b = 2$, $c = 5$

Step 2 : Apply the formula $(a + b)(a + c) = a^2 + ac + ab + bc$

Step 3 : Substitute $x^2 + 5x + 2x + 10$

Step 4 : Collect like terms $\rightarrow x^2 + 7x + 10$

	x	5
x	x^2	$5x$
2	$2x$	10

Tip: Each cell of the grid is the product of its row header and column header.

Expanding vs Factorising — What is the Difference?

EXPANDING	FACTORISING
Open the brackets to reveal all terms	Close the terms back into brackets
$(x + 3)^2 \Rightarrow x^2 + 6x + 9$	$x^2 + 6x + 9 \Rightarrow (x + 3)^2$
$(x + 2)(x + 5) \Rightarrow x^2 + 7x + 10$	$9x^2 - 16 \Rightarrow (3x + 4)(3x - 4)$

PRACTICE TIME!

Solve each question below. Show every step of your working. Use the identity formulae from the lesson — no skipping steps!

Practice 1 · Expand	$(a + 5)^2$	Hint: Identity 1
Working:		
Answer:		

Practice 2 · Factorise	$25x^2 - 9$	Hint: Identity 3
Working:		
Answer:		

Practice 3 · Expand	$(x - 3)(x + 4)$	Hint: Identity 4
Working:		
Answer:		

Practice 4 · Expand	$(2m - 7)^2$	<i>Hint: Identity 2</i>
Working:		
Answer:		

Practice 5 · Factorise	$49y^2 - 36$	<i>Hint: Identity 3</i>
Working:		
Answer:		

Practice 6 · Expand	$(p + 6)(p - 6)$	<i>Hint: Identity 3</i>
Working:		
Answer:		

Key Takeaways

1	Four identities to master $(a+b)^2$, $(a-b)^2$, a^2-b^2 , and $(a+b)(a+c)$
2	Identities save time Recognise the pattern → apply the shortcut → write the answer directly
3	Expanding opens brackets Go from factored form to expanded form: $(a+b)^2 \Rightarrow a^2 + 2ab + b^2$
4	Factorising closes brackets Go from expanded form to factored form: $a^2-b^2 \Rightarrow (a+b)(a-b)$

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Practice makes perfect

The more problems you solve, the faster you will recognise which identity to use

You are an Algebra Star! Keep Practising! ■